

Zubair Mughal

LIVE CODING CLASSES FOR KIDS, TEENS & BEGINNERS

JV

COURSE 07 OF 12
CURRICULUM GUIDE

Java Development for Teens

Object-oriented programming, done properly

GRADE BAND

Grade 9–12

DIFFICULTY

Intermediate

DURATION

16 weeks · 2 sessions/week

ABOUT THIS COURSE

Course Overview

Java teaches the object-oriented discipline used across enterprise software, Android development and most university intro-CS courses — a rigorous complement to the more forgiving languages students start with.

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Prerequisites: Python for Kids & Teens, or equivalent programming basics.

LEARNING OUTCOMES

What Students Will Learn

- Understand Java syntax, compilation and program structure
- Apply object-oriented principles: classes, objects, encapsulation
- Use inheritance and polymorphism to design flexible programs
- Work with arrays, ArrayLists and basic collections
- Handle errors gracefully using exceptions
- Design and build a multi-class object-oriented application

WHAT WE'LL USE

Tools & Technologies

Java (JDK) IntelliJ IDEA / VS Code Git

CURRICULUM

Module-by-Module Breakdown

MODULE	FOCUS
01 · Java Fundamentals & Setup	Syntax, compiling and running programs.
02 · Variables, Types & Control Flow	Java-specific syntax and static typing.

MODULE	FOCUS
03 · Object-Oriented Basics	Classes, objects and constructors.
04 · Encapsulation & Methods	Getters/setters and access modifiers.
05 · Inheritance & Polymorphism	Extending classes and overriding methods.
06 · Collections	Arrays, ArrayLists and iterating over data.
07 · Exception Handling	try/catch and defensive programming.
08 · Capstone OOP Project	Multi-class application design and build.

LEARNING BY BUILDING

Projects Students Will Build

- Library / Inventory Management System — The course's core OOP capstone.
- Simple Banking Simulation — Classes and encapsulation in practice.
- Student Grade Tracker — Collections and applied logic.

BEYOND THE SYNTAX

Skills Developed

OOP design Java syntax Exception handling Software structuring

PORTFOLIO OUTCOME

A multi-class Java application with a class diagram and a README explaining the design.

RECOMMENDED NEXT STEP

DSA for Teens, or SQL for Teens & Beginners